On-Chain Evidence, Off-Chain Adjudication: A Layered Approach to Smart-Contract Dispute Resolution

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Abstract

This paper makes a focused legal argument about smart-contract dispute resolution. The dominant on-chain dispute-resolution designs—Kleros's Schelling-jury arbitration, Aragon Court's DAO-governed jury, optimistic-rollup challenge windows—attempt to perform adjudication on chain. We argue that this is a category error: settlement is on-chain by nature, adjudication is off-chain by nature, and conflating the two re-introduces the discretionary intermediary that smart contracts were supposed to remove.

We develop the argument by reference to a recently developed settlement primitive (mechanism paper), formally verified in the implementation paper, whose three-layer enforcement architecture explicitly separates the bonding game (Layer 1), the peer-coordination subgame (Layer 2), and the evidentiary record that supports off-chain proceedings when the first two layers fail (Layer 3). The contribution of this paper is to develop Layer 3 in detail: the evidentiary properties of the on-chain record, the relationship between those properties and existing electronic-evidence law (eIDAS, UCC §1-201, the Singapore Electronic Transactions Act, the UNCITRAL Model Law on Electronic Commerce), and the jurisdictional implications of decoupling settlement from adjudication.

The argument is not that smart contracts replace courts. It is that smart contracts can produce a form of evidence that courts already know how to use, and that the cleanest division of labor is one in which the protocol does what it does well (deterministic settlement) and the legal system does what it does well (contextual adjudication). The protocol is silent on what counts as "satisfactory performance" or "material breach"—those are adjudicative determinations—but loud on the facts that adjudication needs as input.

Keywords: electronic evidence, smart-contract law, dispute resolution, blockchain evidence, eIDAS, UNCITRAL, jurisdictional flexibility

1 Introduction

Smart-contract dispute resolution is currently a contested design space. The dominant architectures attempt to perform adjudication on chain, typically through some variant of token-weighted juror selection or DAO governance. Kleros [Kleros, 2019] uses Schelling-point juror voting; Aragon Court [Aragon, 2017] embeds arbitration in DAO infrastructure; optimistic mechanisms [Optimism, 2021, UMA, 2020] use challenge windows in which assertions stand unless contested. Each of these approaches addresses the dispute-resolution problem by importing the discretionary actor that would otherwise reside in a court: the juror, the DAO voter, the challenger.

This paper argues that the import is a mistake. The claim is not that on-chain juries are bad mechanisms (they are interesting mechanisms with documented properties); it is that they are solving the wrong problem. The right problem is not "how do we adjudicate disputes on

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chain" but "how do we produce evidence that off-chain forums can use to adjudicate efficiently." The former requires re-creating, in adversarial token-economic settings, the institutional infrastructure (impartial juror selection, jurisdictional fit, due process) that civil legal systems have refined over centuries. The latter requires only that the on-chain artifact meet existing evidentiary standards—a substantially easier problem, and one for which the legal infrastructure already exists.

We make the argument in the context of Figaro, a settlement primitive whose mechanism is established in the mechanism paper and whose implementation is verified in the implementation paper. Figaro explicitly does not attempt on-chain dispute resolution. Its three-layer enforcement architecture treats Layer 1 (bonding) and Layer 2 (peer coordination) as the primary defenses, and Layer 3 (immutable evidence) as the residual mechanism for cases where Layers 1 and 2 are insufficient. The present paper develops Layer 3.

What this paper is not. This paper does not provide legal advice. It does not predict the admissibility of any specific on-chain artifact in any specific jurisdiction, nor does it claim that the on-chain record is itself sufficient for any particular legal claim. It argues for a design discipline (do not perform adjudication on chain) and catalogues the evidentiary properties of the artifact that the discipline produces. Practitioners who wish to rely on those properties in specific proceedings should obtain jurisdiction-specific counsel.

Paper organization. Section 2 states the layered architecture. Section 3 develops the evidentiary properties of the on-chain record. Section 4 compares those properties to existing electronic-evidence law. Section 5 addresses jurisdictional implications. Section 6 argues against on-chain adjudication as a category. Section 7 concludes.

2 The Three-Layer Architecture

Figaro's enforcement architecture is layered. We summarize it here from the legal-design perspective; the mechanism details are in the mechanism and implementation papers.

Layer 1 — bonding. Each party to a transaction locks $2\times$ the transaction value (asymmetric for N-party processes; the seller's bond scales with upstream value). The locked capital is recoverable only through buyer-initiated atomic resolution. Cooperation is the strictly dominant strategy; defection costs the full bond. The vast majority of disputes are prevented at this layer because the incentive to defect is structurally negative.

Layer 2 — peer coordination. Atomic resolution induces a weakest-link subgame among co-sellers in a process tree: any one seller's defection causes all sellers to lose their bonds. Honest sellers therefore have direct economic incentive to monitor and pressure underperforming co-sellers. This layer handles disputes that survive Layer 1 by internalizing them as intra-cohort coordination problems.

Layer 3 — evidence. The on-chain record. Every commitment, every bond deposit, every resolution (or its absence) is recorded with block timestamps and cryptographic signatures. When Layers 1 and 2 fail—when a party acts irrationally, when a co-seller cohort cannot self-coordinate, when external circumstances (regulatory action, force majeure, participant death) prevent normal resolution—Layer 3 produces the artifact that off-chain dispute forums need to adjudicate.

What is and is not on chain. Settlement is on chain. Adjudication is off chain. The split is deliberate. Settlement is deterministic, automated, and public—all properties that on-chain execution provides natively. Adjudication is contextual, discretionary, and jurisdiction-specific—all properties that on-chain execution is poorly suited for. The split lets each layer do what it does best and avoids forcing one to imitate the other.

3 Evidentiary Properties of the On-Chain Record

Before enumerating the evidentiary properties of the on-chain record, we draw a distinction the remainder of this section relies on. The "on-chain record" is not a single artifact class; it is two artifact classes with substantially different evidentiary character.

Class A: settlement byproduct. Commitment payloads, bond deposits, and resolution calls. These are produced as a mechanical side effect of the bonding game: no party can collect the settlement payout without the corresponding on-chain events having been emitted. Their integrity as evidence is underwritten by the game itself — an actor cannot opt out of producing them without opting out of settlement.

Class B: discretionary attestation. Schema-typed lifecycle events, disclosure records, proximity witnesses, and other party-generated attestations emitted by the attestation coordinator. These are voluntary: a party chooses whether to attest, when to attest, and what content to place in the attestation. The schema validator ensures the content is well-formed; it does not and cannot ensure the content is accurate. A party who selectively attests (silence about underperformance, precise attestation of performance) can produce a Class B record that survives authentication but misrepresents reality.

The six properties enumerated below apply *unconditionally* to Class A and *conditionally* to Class B: each Class B attestation carries the integrity of the signature and timestamp (Class A characteristics), but the accuracy of its content depends on the attestor's incentive structure at the moment of attestation. A cryptoeconomics reader will recognize Class B as subject to the same self-reporting incentive problems that attach to every voluntary disclosure; adjudication should treat Class B under ordinary authentication-and-hearsay doctrine rather than extending Class A's byproduct assurance to it. An attestation-bond layer (stake posted per attestation, slashable on proven falsity) is a plausible Class B strengthening but lies outside the kernel and is not yet developed in the protocol.

Evidentiary completeness and constitutive primacy. A further distinction matters for certain downstream uses: the on-chain record is not merely *admissible* as evidence but *constitutive* of the accounting record of the process. The companion accounting paper develops the argument that a FIGARO process is a self-closing ledger period in the precise accounting sense — the custodial balance of the contract is the balance-sheet position; commit and resolve events are journal and closing entries; the process closes when resolution discharges all custodial liabilities. For disputes whose subject matter is the accounting record of economic activity settled through the protocol, the on-chain record is not corroborating evidence for a party-held book; the chain is the book, and any party-held document is a summary or translation of it. The admissibility analysis of the present section applies to the record's evidentiary weight in proceedings; the constitutive analysis applies to the record's primacy within the economic activity it documents.

The on-chain record produced by FIGARO has six properties that together constitute its evidentiary value.

(1) Timestamped. Every state transition is associated with a block timestamp. The timestamp is set by block proposers under consensus rules; it is not subject to retroactive modification. For practical purposes, the timestamp on a finalized block is a qualified time stamp in the eIDAS sense.

(2) Cryptographically signed. Every commitment is signed by both parties using EIP-712 typed structured data [EIP-712, 2017]. The signatures bind the commitment content (payment, parties, denomination, agreement hash, deadline) to the signers' private keys. Recovery is deterministic; tampering with any field invalidates the signature.

(3) Content-addressed. Order identifiers are content hashes (orderHash = keccak256(processId ‖ structHash
The same commitment content always produces the same identifier; different content always produces different identifiers (collision-resistant under the assumed hash function). This makes the record unforgeable in a specific sense: no second commitment can claim the same identifier.

(4) Immutable. On a finalized block, the record cannot be modified or deleted. A party seeking to repudiate a commitment cannot purge the chain of the relevant log entries. This is a stronger property than what electronic records normally enjoy.

(5) Publicly verifiable. Anyone can read the chain and replay the verification logic. The evidence is not held by any single party; it is held by the network. Disputing parties therefore cannot disagree about what the record says—they can disagree only about what the record *means*, which is precisely the question adjudication is suited to.

(6) Contains commitment intent. The signed payload includes a content hash h of the off-chain agreement. The off-chain agreement itself can be anything the parties choose—a structured contract, a free-text PDF, a schema-typed JSON document. The on-chain record does not interpret the agreement; it commits both parties to a specific document via the hash. A dispute over performance is then a dispute over the referenced document, with the on-chain hash establishing *which* document is at issue.

These six properties, combined, produce an artifact that is substantially *stronger* than the electronic records most courts already accept. Courts admit emails, electronic signatures, and database extracts as evidence subject to authentication challenges; the on-chain record satisfies many authentication requirements at the cryptographic level rather than at the custodial level.

4 Legal Fit: Existing Electronic-Evidence Frameworks

We sketch how the six properties above interact with four major electronic-evidence frameworks. The treatment is illustrative; a full doctrinal analysis is jurisdiction-by-jurisdiction.

eIDAS (EU Regulation 910/2014). eIDAS distinguishes electronic signatures, advanced electronic signatures (AdES), and qualified electronic signatures (QES). EIP-712 signatures meet the AdES technical requirements (uniquely linked to the signatory; capable of identifying the signatory; created using means under the signatory's sole control; linked to the signed data such that any subsequent change is detectable). They do not, by themselves, meet the QES requirements (which include a qualified certificate from a trust service provider). However, the AdES status is sufficient for most contractual purposes, and the eIDAS framework explicitly provides that the legal effect of an electronic signature shall not be denied solely because it is in electronic form. eIDAS also defines qualified electronic time stamps; block timestamps on chains with public consensus rules satisfy the technical requirements but typically not the trust-service-provider designation. The practical effect: the on-chain record is admissible electronic evidence under eIDAS, with weight determined by the court's contextual assessment.

UCC §1-201 (US). The Uniform Commercial Code defines "record" as "information that is inscribed on a tangible medium or that is stored in an electronic or other medium and is retrievable in perceivable form." On-chain log entries satisfy this definition without ambiguity. UCC §1-201 also defines "signature" inclusively ("any symbol executed or adopted with present intention to adopt or accept a writing"); EIP-712 signatures, being intentionally executed by the signer's private key on the typed commitment, qualify. The Uniform Electronic Transactions Act (UETA) and the federal E-SIGN Act extend the same principle to contracts more broadly. The practical effect: on-chain commitments are records and signatures under US commercial law.

Singapore Electronic Transactions Act (Cap. 88). Singapore's ETA closely follows the UNCITRAL Model Law and provides functional equivalents for "writing", "signature", and "original" in electronic form. The ETA was amended in 2021 to recognize electronic transferable records, extending its scope to instruments of title. On-chain records satisfy the writing and signature requirements directly; the transferable-record provisions are particularly relevant for process trees in which the right to receive performance moves along the supply chain.

UNCITRAL Model Law on Electronic Commerce. Adopted in 1996 and supplemented by the 2017 Model Law on Electronic Transferable Records, the UNCITRAL framework articulates the functional-equivalence principle that underlies most modern electronic-evidence regimes worldwide. The Model Law's three core requirements—accessibility for subsequent reference, integrity of the information, and identifiability of the originator—are all satisfied by on-chain commitments. The practical effect: jurisdictions that have adopted UNCITRAL-derived electronic-transactions law (over 80 as of 2024) provide an established framework for treating on-chain commitments as admissible electronic evidence.

The summary picture. The on-chain record is admissible electronic evidence in every jurisdiction whose electronic-evidence law derives from the UNCITRAL framework or implements equivalent functional principles. What varies across jurisdictions is the weight courts assign to the evidence (a question of judicial practice rather than admissibility) and the further procedural requirements for authenticating the chain itself (typically satisfied by expert testimony or by the court taking judicial notice of well-known chains). No jurisdiction we are aware of categorically excludes on-chain records from evidentiary consideration.

5 Jurisdictional Implications

The decoupling of settlement from adjudication has substantive jurisdictional consequences.

Forum is contractual. The on-chain commitment includes an off-chain agreement hash h. That off-chain agreement can—and typically should—include a forum-selection clause, a choice-of-law clause, and an arbitration clause if the parties prefer arbitration to court. The on-chain record proves what was agreed; the off-chain agreement specifies where and under what law the agreement is adjudicated. Parties who omit a forum clause default to whatever private-international-law analysis the seised forum applies; this is a known regime, not a novel problem.

Forum-shopping is bounded. A party seeking to forum-shop is bounded by (a) the forum-selection clause in the off-chain agreement, (b) the seising forum's own jurisdictional rules, and (c) the recognition-and-enforcement treaties between the seising forum and any forum where the counterparty's assets reside. None of these constraints is weakened by the on-chain settlement; they all operate on the off-chain agreement and the parties' real-world circumstances.

Evidence is jurisdiction-displaced. The integrity of the on-chain evidence does not depend on the seising forum's recognition of any specific cryptographic standard. The signatures verify, or they do not; the chain hashes match, or they do not. A forum that doubts the cryptographic analysis can engage an expert; a forum that does not is in the position of any forum facing forensic evidence.

Cross-border processes. A process tree can involve parties in multiple jurisdictions simultaneously. The on-chain settlement is jurisdiction-blind: the bond mechanics work identically regardless of where the parties reside. Off-chain disputes that arise from such processes are handled by the same private-international-law machinery that already handles cross-border disputes in conventional commerce—typically with the on-chain record providing substantially more evidentiary clarity than the parties would have produced under conventional bookkeeping.

6 Why Not On-Chain Adjudication

We have argued the positive case for off-chain adjudication. We now argue the negative case against on-chain adjudication. Several patterns occupy this space; we treat each briefly.

Schelling-point juror selection (Kleros). A token-weighted juror panel votes on the disputed question; the plurality wins; jurors who voted with the plurality are rewarded, jurors who voted against are slashed. The mechanism is elegant in its abstraction but has three problems for serious adjudication. First, the juror panel is not selected for jurisdictional fit—a Kleros juror in jurisdiction A may know nothing of jurisdiction B's law that the parties agreed should govern. Second, the juror's incentive is to vote with the perceived plurality, not to vote for the legally correct outcome—a Schelling point on plurality is structurally distinct from a Schelling point on truth. Third, the slashing mechanism imports a new dispute (was the slashing fair?) that has no terminal forum on chain.

DAO-governed arbitration (Aragon Court). Aragon Court is structurally similar to Kleros, with the juror-selection layer integrated with DAO governance. This inherits all of Kleros's problems plus the additional problem that DAO voting power is plutocratic in proportion to token holdings; the wealthy juror has structurally more votes than the poor juror. Whether or not one views this as a desirable property of arbitration is a normative question; that it differs from the conventional "one juror, one vote" premise of jury trials is a doctrinal question that has not been settled.

Optimistic challenge windows. Optimistic mechanisms [Optimism, 2021, UMA, 2020] avoid the juror problem by allowing assertions to stand unless challenged within a fixed window. This works for settlement-layer state machines but does not work for adjudication, because adjudication is precisely the activity of assessing an assertion that has been challenged. An optimistic "adjudication" is just an assertion with a deadline—it has no mechanism for resolving the underlying question, only for resolving the question of who blinked first.

The structural objection. All on-chain adjudication mechanisms share the same structural flaw. The kernel mechanism of the mechanism paper establishes that any unilateral exit from the bonded state requires either a third-party decision (which introduces an unbonded actor whose incentives are not constrained by the bond structure) or a payoff modification that weakens the Nash equilibrium. On-chain adjudication is the third-party-decision case: the juror, the DAO voter, the challenger is the unbonded actor whose decision determines the outcome. Whatever the mechanism designer does to constrain that actor (token slashing, Schelling incentives, reputation), the actor is still unbonded and the kernel's self-enforcing property is broken.

The off-chain alternative. Off-chain adjudication by an existing legal forum has the opposite structure. The forum's incentives are constrained by its institutional context—judges face appellate review, arbitrators face institutional reputation, both face professional sanction. The forum is not unbonded in the kernel sense; it operates inside a different bond structure (its institutional authority). The off-chain evidence record is therefore consumed by an actor whose incentives are already constrained by a regime that has been refined for centuries. The protocol does not re-implement that regime; it provides input to it.

Composition with Kleros as forum, not as kernel adjudicator. The objection above is to on-chain adjudication *at the kernel layer*. It is not an objection to Kleros, or to any other adjudication mechanism, when the parties' agreement designates it as the forum. The kernel of FIGARO is forum-agnostic: it does not adjudicate; it produces an evidence record. If two parties sign an off-chain agreement whose arbitration clause names Kleros as the dispute forum, the evidence record produced by their FIGARO commitments is exported to Kleros and the dispute is adjudicated there. From the kernel's perspective Kleros is the off-chain forum the parties chose, in the same sense that the Singapore International Arbitration Centre or the New York Supreme Court can be the off-chain forum the parties choose. The structural critique above is not an endorsement of some forums over others; any forum the parties name in their agreement is their forum to choose, and FIGARO composes with it by exporting evidence to it.

Engagement with the legal-academic literature on Kleros. A small but real legal-academic literature has developed around Kleros and its peers. Ortolani [2019] is the principal engagement from the international-arbitration side, raising the recognition-and-enforcement question under the New York Convention [New York Convention, 1958]: can a Kleros decision be enforced abroad as an arbitral award, and on what doctrinal basis? Kaal [2020] reviews the broader landscape of decentralized dispute resolution. De Filippi and Wright [2018] situates Kleros and similar mechanisms within the lex cryptographia frame. Rabinovich-Einy and Katsh [2017] place them in the longer trajectory of online dispute resolution. The literature is not yet large but it is real, and its general posture is compatible with this paper's: most of the legal-academic engagement with Kleros expresses concern about adjudicative-quality and recognition issues, not about the existence of the mechanism. The recognition-and-enforcement question raised by Ortolani and others is in our reading the binding legal-doctrinal question, and the layered architecture of Section 2 mitigates it: the on-chain record is admissible electronic evidence under the frameworks of Section 4 regardless of how the forum that consumes the evidence is constituted, and forum-clause discipline allows the parties to name a forum (Kleros, SIAC, court) whose decisions they have a clean recognition theory for in the jurisdiction of intended enforcement.

7 Conclusion

We have argued for a layered approach to smart-contract dispute resolution: settlement on chain, adjudication off chain, with a deliberately designed evidentiary record that bridges the two. The architecture inherits the strengths of each layer (chain determinism for settlement, legal-system contextual judgment for adjudication) without forcing either to imitate the other. It also resolves the structural objection to on-chain adjudication: the unbonded juror does not need to exist because the legally constituted forum already does.

The contribution is modest but, we believe, important. The debate over smart-contract dispute resolution has been dominated by the assumption that on-chain settlement implies on-chain adjudication. The decoupling we propose is a different design posture, and it permits a substantially cleaner alignment with existing electronic-evidence law.

What remains for legal scholarship. Three questions are out of scope for this paper but warrant sustained legal-scholarly attention. First, the doctrinal analysis of on-chain commitments under specific jurisdictions' contract-formation, evidence, and electronic-signature laws is substantial and necessarily jurisdiction-by-jurisdiction. Second, the question of how courts should address chain-finality assumptions (what happens when the underlying chain reorganizes substantially after a contested transaction) requires technical-legal collaboration that has barely begun. Third, the question of whether and when the on-chain record should be treated as self-authenticating (in the US Federal Rules of Evidence sense, or its analogues in other common-law jurisdictions) is a question of judicial practice that will develop with case law.

What remains for protocol design. On the protocol side, the principal open question is how schema-typed attestations (implementation paper) interact with off-chain dispute proceedings. The attestation surface allows parties to record structured evidence (geo proofs, lifecycle stage transitions, GHG disclosures) on chain in addition to the bare commitment. How that structured evidence is treated by off-chain forums—whether as supplementary fact, as binding stipulation, or as expert testimony—is design space that has not yet been explored.

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